

TEJESWARISAI KOLLI

Data & AI Engineer

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PROFESSIONAL SUMMARY

Data & AI Engineer with 4 years of professional experience in cloud data engineering and 2 years of hands-on experience in Generative AI. Skilled in designing and delivering scalable data platforms, ETL/ELT pipelines, data warehousing solutions, real-time data processing, and AI-powered applications. Proficient in Python, SQL, PySpark, Apache Beam, Apache Airflow, BigQuery, Dataflow, Cloud Composer, Cloud Spanner, Databricks, and Azure Databricks. Experienced in developing enterprise AI solutions using LLMs, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), GPT-4o, ChromaDB, FastAPI, and Streamlit. Strong understanding of modern cloud and AI ecosystems, including Microsoft Azure services such as Azure OpenAI, Azure AI Studio, Azure AI Search, Azure Data Factory, Azure Synapse Analytics, and Azure Data Lake. Proven ability to build intelligent data solutions, multi-agent AI systems, and business-focused analytics platforms while collaborating with cross-functional stakeholders in Agile environments.

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Prodapt Solutions Pvt Ltd

Client: Verizon

Project: One Network Data *Jan 2026 – Present*

- Engineered and managed Airflow DAGs in Cloud Composer to orchestrate batch and streaming ETL/ELT pipelines, improving data reliability and operational efficiency.
- Developed scalable data processing solutions using Apache Beam and Dataflow to ingest, transform, and load large-scale enterprise data into Cloud Spanner and BigQuery.
- Designed and optimized cloud-based data warehousing solutions in BigQuery, enabling analytics and business intelligence reporting.
- Built an AI-powered natural language query assistant using LangChain, GPT-4o, and BigQuery, enabling business users to retrieve insights without SQL expertise.
- Developed a RAG-based data lineage and observability solution using ChromaDB and prompt engineering to improve data quality monitoring and anomaly detection.
- Collaborated with cross-functional stakeholders to gather requirements, design data solutions, and support data modernization initiatives across the enterprise.

Project: Message Code Simplification *July 2025 – Dec 2025*

- Architected and developed an enterprise Consumer Data Platform (CDP) on GCP, integrating customer data from multiple source systems into a unified customer data model.
- Designed and implemented scalable real-time and batch ETL pipelines using Apache Kafka, Dataflow, Python, and SQL for large-scale customer data processing.
- Developed data transformation, validation, and enrichment frameworks to improve data quality, governance, and consistency across downstream systems.
- Built and optimized BigQuery data marts, views, and aggregations supporting analytics, reporting, and machine learning use cases.
- Partnered with business and analytics teams to translate requirements into scalable data engineering solutions and actionable insights.
- Implemented data governance and auditing controls throughout the data lifecycle, ensuring compliance and data reliability.

- Enabled customer activation and personalized marketing by delivering curated datasets to Adobe Experience Platform and Adobe Campaign

GenAI Training

May 2026 – Jun 2026

Capstone Project: SGEIA (Smart Grid Energy Intelligence Assistant)

- Completed Prodapt's GenAI program covering Prompt Engineering, RAG, LangChain, LangGraph, multi-agent systems, Guardrails, DeepEval/Ragas evaluation, LangSmith/Langfuse observability, and FastAPI development.
- Built SGEIA using LangGraph StateGraph with GPT-4o, ChromaDB (hybrid BM25 + semantic RAG), XGBoost for grid stability prediction, and Isolation Forest for smart meter anomaly detection.
- Delivered a production-ready SGEIA system with FastAPI streaming backend and Streamlit dashboard, achieving >0.85 faithfulness score via DeepEval quality gates and full LangSmith observability across 5 agent nodes.

GCP Data Engineer

Proxima Systems Pvt Ltd

Client: SnippetSentry

Project: SnippetSentry May 2022 – Dec 2024

- Designed and delivered end-to-end data pipelines on GCP using Apache Airflow, BigQuery, and Cloud Storage, supporting high-frequency ingestion and analytics reporting.
- Managed Cloud Storage buckets, IAM roles, and access controls to maintain data security, compliance, and auditability across environments.
- Built and optimised 15+ BigQuery datasets, tables, and SQL queries, cutting average query execution time by 40% and improving analytics turnaround.
- Proactively monitored pipeline health, identified bottlenecks, and resolved failures — significantly reducing latency and improving data reliability.
- Authored runbooks, data flow diagrams, and test coverage for 10+ deployed pipelines, enabling faster onboarding and quicker incident resolution.
- Mentored new team members on Apache Airflow best practices, standardising development workflows and accelerating team productivity.

Project: AI Flow Mapper

Dec 2023 – Dec 2024

- Developed AI Flow Mapper, reducing manual diagram creation time by ~60% by automatically transforming requirement keywords and JSON inputs into structured mind maps and flow diagrams.
- Designed the data ingestion layer to parse and normalize JSON inputs, enabling dynamic generation of structured diagram outputs.
- Built backend GCP pipelines for transformation logic, storing intermediate results in Cloud Storage and serving outputs via BigQuery.
- Reduced requirements-to-diagram turnaround by 3x, enabling product and engineering teams to visualise complex workflows in minutes rather than hours.

Project: Bentley Systems

Feb 2022 – April 2022

Client: Bentley | Role: SQL Developer

- Wrote and optimized SQL queries to extract, manipulate, and analyse data from relational databases.
- Developed complex joins, subqueries, and aggregate functions to fulfil business reporting needs.
- Designed database schemas including tables, primary and foreign keys, and indexes.
- Performed data cleaning and validation on 5+ datasets, resolving inconsistencies and improving data accuracy for downstream reporting.
- Supported reporting dashboards by writing backend SQL queries for data sourcing and summaries.

TECHNICAL SKILLS

Cloud Platforms	Google Cloud Platform (BigQuery, Cloud Storage, Dataflow, DataProc, Pub/Sub, Cloud Spanner, Cloud Composer) Microsoft Azure (Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, Azure Machine Learning, Azure SQL)
Data Engineering & Analytics	ETL/ELT Development, Data Warehousing, Data Migration, Data Modernization, Real-Time Data Processing, Data Modeling, Data Quality & Governance, Real-Time Data Processing, Analytics Engineering
Data Tools	Apache Airflow, Apache Beam, Apache Spark, Databricks, MongoDB
AI / GenAI	LangGraph, LangChain, OpenAI GPT-4o, Gemini, ChromaDB, RAG, Prompt Engineering, LangSmith, DeepEval, Ragas, Langfuse
AI Serving	FastAPI (SSE Streaming), Streamlit, uvicorn
ML & Evaluation	XGBoost, Isolation Forest, SHAP
BI & Automation	Power BI, Google Cloud Console, Looker
Programming	Python, SQL, JavaScript, PySpark
Others	Agile, GIT Labs, JIRA

CERTIFICATIONS

- Python (Core and Advanced) — National Skill Development Corporation
- Introduction to Artificial Intelligence — LinkedIn
- DataBricks Lakehouse - Coursera
- GenAI Program — Prodapt Solutions

EDUCATION

Bachelor of Technology – Electronics & Communication Engineering

Andhra University, Visakhapatnam

2018 – 2022

Intermediate (MPC)

Sri Chaitanya Junior College, Visakhapatnam

2016 – 2018